Recurring decimals

1 Write the following recurring decimals as a fraction in simplest form:

a $x = 0.\dot{1}$ b $x = 0.\dot{3}$ c $x = 0.\overline{6}$ d $x = 2.\dot{3}$

e $x = 1.\overline{7}$ f $x = 3.\dot{6}$ g $x = -0.\dot{8}$

2 Write the following recurring decimals as a fraction in simplest form:

a $x = 0.\dot{7}\dot{5}$ b $x = 0.\overline{45}$ c $x = 9.\dot{8}\dot{3}$ d $x = 0.\dot{3}7\dot{9}$

e $x = 0.\overline{216}$ f $x = 7.\dot{6}8\dot{4}$ g $x = 0.\overline{25748}$

3 Write the following recurring decimals as a fraction in simplest form:

a $x = 0.6\dot{3}$ b $x = 0.6\dot{8}$ c $x = 0.0\overline{75}$ d $x = 0.6\dot{8}\dot{7}$

e $x = 0.76\dot{2}$ f $x = 0.00\overline{42}$ g $x = 9.00\overline{47}$

4 Express $3.9 \div 9$ as:

a A recurring decimal. b A simplified fraction.